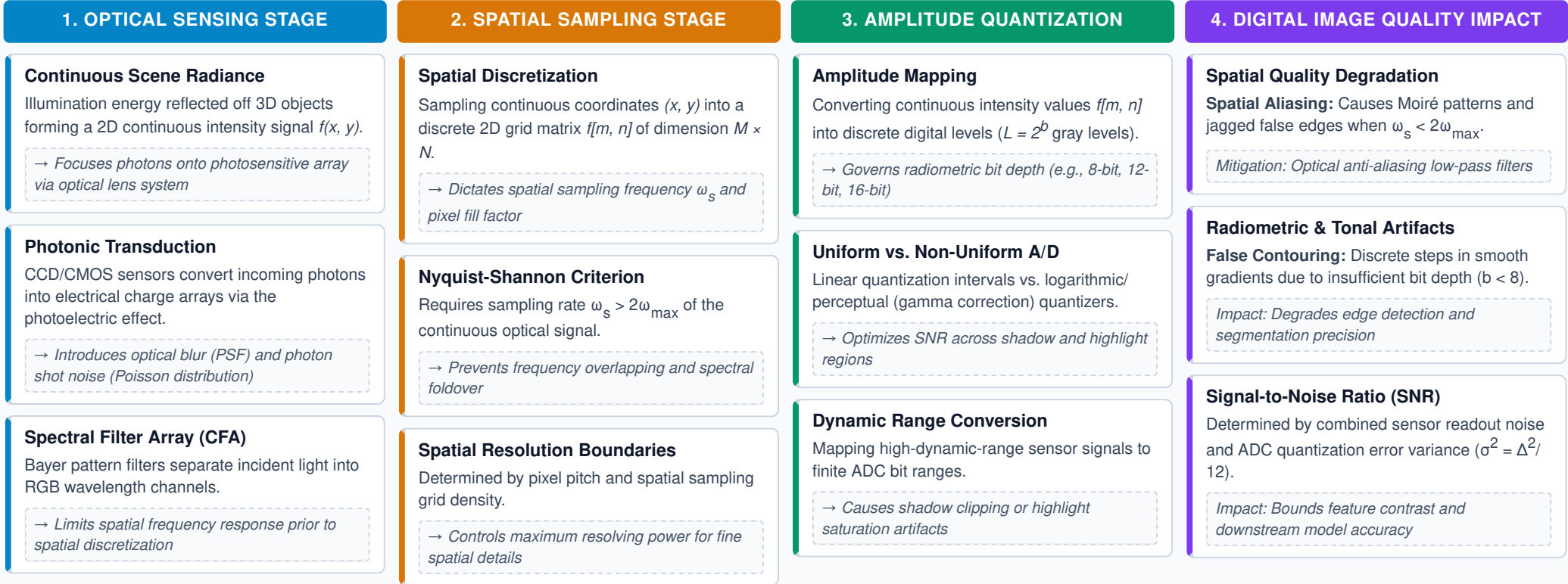


Workflow of Digital Image Formation: Sensing, Sampling, and Quantization

A systematic concept map mapping continuous real-world radiance to discrete digital representations and evaluating downstream image quality trade-offs



Empirical Literature Synthesis & Foundational Principles

WORKFLOW STAGE	KEY THEORETICAL CONCEPT	INTERCONNECTION & TECHNICAL MECHANISM	IMPACT ON DOWNSTREAM IMAGE QUALITY	FOUNDATIONAL REFERENCE
Sensing	Point Spread Function (PSF) & MTF	Lens diffraction and pixel integration act as spatial low-pass filters prior to sampling.	Attenuates high spatial frequencies, reducing sharpness and fine edge contrast.	Goodman (2005) / Holst (2011)
Sampling	Nyquist Sampling & Aliasing	Undersampling spectral frequencies above $\omega_s/2$ folds high frequencies into lower bands.	Generates false Moiré fringe patterns and irreversible spatial distortions.	Shannon (1949) / Gonzalez & Woods (2018)
Quantization	Dynamic Range & Bit Depth (b)	Quantization error model ($e = f - \hat{f}$) introduces additive white noise with variance $\Delta^2/12$.	Low bit depth ($b < 6$) produces false contouring; high bit depth (≥ 12 -bit) preserves fine shadow detail.	Gray & Neuhoﬀ (1998) / Jain (1989)
Quality Trade-off	Spatial vs. Bit Depth Allocation	Total storage/bandwidth $N = M \times N \times b$ trades off spatial grid density against gray-level precision.		Huang (1965) / Pratt (2007)

Workflow Stage	Key Theoretical Concept	Interconnection & Technical Mechanism	Impact on Downstream Image Quality	Foundational Reference
			For detailed textures, higher spatial grid density (M, N) provides higher perceptual quality than excessive bit depth.	

Workflow Synthesis: Digital image formation is an information-losing transformation. Optical sensing establishes the physical SNR and bandwidth ceiling; spatial sampling determines structural resolving power (bounded by Nyquist limits); and quantization dictates amplitude fidelity. Optimal digital image representation requires balancing spatial sampling density and radiometric bit depth under total storage constraints.